

CHECKLIST
SMALL CRAFT - STEERING GEAR - CABLE OVER PULLY SYSTEMS
Ref.: EN ISO 8847:2021

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:	
Watercraft model name:	



	Subject to check	Clause	Requirements	Checked ?
1	Craft equipped with outboard engine(s) up to and including 37 kW total power?	1	[Yes / NA ?]	
2	All components are fastened securely to the structure of the boat, reinforced where necessary, especially at the steering mechanism bulkhead/pedestal mounting and at pulleys.	4.1	[Yes ?]	
3	For sailing boats: The steering arm connection to the rudder shaft is capable of transmitting the steering torque to the rudder.	4.2	[Yes / NA ?]	
4	Cable over pully steering system is not used to stear craft having outboard engine(s) in excess of 37 kW power.	4.3	[Yes ?]	
5	Threaded fasteners whose integrety affects safe operation of the steering system are provided with a locking means.	4.4	[Yes / NA ?]	
6	Threaded fasteners that can be expected to be disturbed by installation or adjustment procedures are locked by a device or incorporate an integral locking means.	4.4.1	[Yes / NA ?]	
7	Loose lock washers, distorted thread nuts or seperatly applied adhesives are not used.	4.5	[Yes ?]	
8	Components of the steering system are accessible.	4.10	[Yes ?]	
9	Steering system requirements			
10	Outboard engine: Steering stops permit at least 30° of angular movement either side of centre.	5.1	[Yes / NA ?]	
11	Steering wheels and helm shafts fit each other.	5.3	[Yes ?]	
12	Label: Permanently marking of the steering mechanism with largest diamer and deepest dish steering wheel. Affixed: Visible on steering mechanism, even when wheel is removed.	5.3.1	[Yes ?]	
13	All steering components are of sufficient strength.	5.3.2	[Yes ?]	
14	Cables used for steering system on sailing boats:			
15	- are flexible.	5.5.1	[Yes / NA ?]	
16	- meet the wire characteristics given in ISO 2408:2017.	5.5.1	[Yes / NA ?]	
17	- the wire rope tension is adjustable by means of an appropriate tensioner.	5.5.1	[Yes / NA ?]	
18	- are aligned with the grooves when leaving a steering arm.	5.5.1	[Yes / NA ?]	
19	- the cable termination is appropriate to transfer the cable load.	5.5.1	[Yes / NA ?]	
20	Cables used on steering system of craft with outboard engine:			
21	- are flexible.	5.5.2	[Yes / NA ?]	
22	- meet the wire characteristics given in ISO 2408:2017.	5.5.2	[Yes / NA ?]	
23	Pulleys used for steering system on sailing boats:			

compliant: Yes or V

not applicable: NA

follow up on variation report: Rpt or X

24	- pulley thread diameter and groove width suitable for cable used.	5.6.1	[Yes / NA ?]	
25	- means to secure the swivel pulley in the correct position.	5.6.1	[Yes / NA ?]	
26	- pulleys are fitted with a means to guide the cables into and out of the sheaves.	5.6.1	[Yes / NA ?]	
27	Pulleys used on steering system of craft with outboard engine:			
28	- the steering cable pulleys shall be located so that the cable comes off the steering cable drum at $90^\circ \pm 5^\circ$ to the steering axis at the centre of travel. Clear space shall be provided behind the dash or pedestal to route the steering cables to pulleys.	5.6.2	[Yes / NA ?]	
29	- angle formed between the pulley pin axis and the axis of both cable are $90^\circ \pm 5^\circ$.	5.6.2	[Yes / NA ?]	
30	- all pulleys, mounting and cable straps are bolted or solid riveted.	5.6.2	[Yes / NA ?]	
31	- no swivel pulley are used at the outboard engine connection or where the pulley changes the cable direction more than 135° .	5.6.2	[Yes / NA ?]	
32	Conduit used on sailing boat systems:			
33	Conduit used on sailing boat steering system are of sufficient flexibility to permit a minimum bend radius of 50 times the smallest diameter of cable for use.	5.7	[Yes / NA ?]	
34	Conduit used on sailing boat steering system are finished at each end with a conduit fitting.	5.7	[Yes / NA ?]	
35	Cable drums used on steering systems of craft with outboard engine have a minimum cable travel of 760 mm.	5.8	[Yes / NA ?]	
36	Cable and pulley steering system			
37	Cable routing openings in the craft structure whose edges can contact the cable during operation have anti-chafe protection.	6.1	[Yes / NA ?]	
38	Clearance provisions to operate free of interference.	6.1	[Yes / NA ?]	
39	There is no open S-hook used.	6.1	[Yes / NA ?]	
40	Sailing boat steering application with or without propulsion engine(s)			
41	The position of the steering stops consider the possibility of dynamic overloading of the system due to shock.	6.2	[Yes / NA ?]	
42	Only the rudder stop(s) shall limit the rotation of the steering system.	6.2	[Yes / NA ?]	
43	The steering arm rotation is installed with one or two stops, suitable fixed to the structure of the craft to limit over-rotation of the steering arm.	6.2	[Yes / NA ?]	
44	Steering systems of craft with outboard engine(s):			
45	The pulleys, mountings and cable end straps are bolted or solid-riveted with at least 5 mm diameter fasteners.	6.3	[Yes / NA ?]	
46	The pulleys, fairleads and end fittings are mounted as indicated in Figure 8 and Figure 9 in the standard.	6.3	[Yes / NA ?]	
47	The craft structure is reinforced at the component mounting fasteners to withstand a minimum force of 3340 N.	6.3	[Yes / NA ?]	
48	Rigging provide a minimum of 760 mm of cable travel. The change of angle in cable at any fairlead does not exceed 5° .	6.3	[Yes / NA ?]	
49	The pulleys and end fittings for the cable clamps are located on the transom face as far from the engine as practicable.	6.3	[Yes / NA ?]	
50	Spring(s) are compressed to one-half their relaxed length after connection of the steering system.	6.3	[Yes / NA ?]	
51	Each spring shall have a minimum of 50 mm of travel and a minimum spring rate of 8,8 N/mm (50 lb/in) to prevent engine shimmy.	6.3	[Yes / NA ?]	
52	Steering cable clearance requirements are determined according to the standard.	6.3	[Yes / NA ?]	

The following questions shall be filled in by the watercraft manufacturer and appropriate documentation shall be submitted to the inspector for verification.

Subject to check	Clause	Requirements	Checked ?
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53	Plastics and elastomers are resistant against salt water, hydrocarbons, Ozone and ultraviolet radiation.	4.7	[Yes ?]	
54	Components of the steering system are resistant to corrosion and galvanically compatible with adjoining components	4.8	[Yes ?]	
55	Components are sized to prevent derailling or jamming of the cables.	4.9	[Yes ?]	
56	Materials used does not effect accuracy and reliability of compasses or navigational instruments, whatever the steering angle may be.	4.11	[Yes ?]	
57	Radius of the steering arm is designed that the cable load is less than 25 % of the cable breaking strength.	5.4	[Yes / NA ?]	
58	The arc generated by the steering arm radius shall be concentric with the rudder shaft centre. The radius at the end of the groove where the cable is led out shall be at least 5 times the diameter of the cable used.	5.4	[Yes / NA ?]	
59	The components of the steering system on a sailing boat transmit the tangential force of 450 N at the standard diameter for the application of loads on the steering wheel or at the maximum torque necessary to steer the rudder without loss of steering function.	7.1.1	[Yes / NA ?]	
60	The components of the steering system of outboard engine(s) withstand 100 000 cycles of operation through 200 mm of cable travel in each direction under a cable load of 110 N without loss of function.	7.1.2	[Yes / NA ?]	
61	Covered steering cables used in outboard engine cable over pulley steering systems have a breaking strength of not less than 4000 N.	7.1.2	[Yes / NA ?]	
62	Covered steering cables used in outboard engine cable over pulley steering systems as installed, the cable covering shall not crack or otherwise expose the cable core under a cable load of 4000 N for 1 min.	7.1.2	[Yes / NA ?]	
63	Covered steering cables used in outboard engine cable over pulley steering systems, the cable covering not strip from the core of a 130 mm sample under a static load of 450 N for 15 min.	7.1.2	[Yes / NA ?]	
64	Outboard engine pulley assemblies are capable of sustaining a 3340 N pulley pin load without permanent deformation.	7.1.2	[Yes / NA ?]	
65	As installed tests - The steering system as installed in a sailing craft pass the axial load test.	7.2.1	[Yes / NA ?]	
66	As installed tests - The steering system as installed in a sailing craft pass the tangential load test.	7.2.1	[Yes / NA ?]	
67	As installed test - The steering system in a craft with outboard engine(s), the cable over pulley system withstand a static cable tension load of 1668 N.	7.2.2	[Yes / NA ?]	
68	As installed test - The steering system in a craft with outboard engine(s), the cable over pulley system withstand a single tangential load in either direction of 267 N.	7.2.2	[Yes / NA ?]	
69	As installed test - The steering system in a craft with outboard engine(s) pass a separte single axial load of 670 N push-pull force.	7.2.2	[Yes / NA ?]	
70	Test reports are submitted to this document.	7	[Yes ?]	

Instructions/Warnings to be included in the owner's manual

71	Routine maintainance procedures.	8	[Yes ?]	
72	Specific precautions critical for correct operation, if relevant.	8	[Yes / NA ?]	
73	Reference to Installation and Instruction manual.	9	[Yes ?]	
74	Largest diameter and deepest dish steering wheel that can be used.	8	[Yes ?]	

Comments:
